CASE STUDY | REAL CLIENT PROJECT

Haq Electronics

An online store and admin panel for an electronics business that distributes electronics and spare parts to markets in Peshawar, Lahore and Karachi.

CLIENT	Haq Electronics, Peshawar
PROJECT	Full-stack e-commerce website with a private admin panel
MY ROLE	Full-stack developer
BUILT WITH	Next.js, React, Node.js, Express.js, MongoDB, JWT and cookies, Cloudinary, Axios, bcrypt, Helmet, CORS
LIVE SITE	www.haq1.com

The client

Haq Electronics is an electronics brand that sells electronics and spare parts in large quantities, including to other sellers. The business ships stock in containers from China and distributes it to markets in Peshawar, Lahore and Karachi.

The problem

Haq Electronics had no online presence, and taking orders was a struggle. Because the business deals in such large volumes, it needed a dependable way for people to see its full range, and a way for the team to keep that range up to date themselves.

What I did

- **Built the store.** The catalogue is organised into clear categories (LED lights, cables, irons, solar inverters, rods and LED spare parts), with product search and filtering. The layout is fully responsive, so it works on phones.
- **Built a private admin panel.** The team logs in to create, edit and delete products, categories and orders. Admin pages are protected, so customers cannot reach them.
- **Secured accounts and data.** Login uses JWT tokens stored in cookies, with role-based protection for admin and customer routes. Passwords are hashed with bcrypt, and the API uses Helmet and CORS.
- **Handled product images properly.** Images are stored on Cloudinary, so the team can upload photos without slowing the site down.
- **Built the back end.** A Node.js and Express.js API with a MongoDB database, connected to a Next.js and React front end.
- **Gave clear feedback.** Actions such as saving a product show notifications, so users always know what happened.

The challenge

This was my first real-world project. Until then I had only built practice and concept projects. The hardest part was not one piece of code. It was bringing together everything I had learned and turning a real client's requirements into a working product, while communicating with him along the way.

The result

Haq Electronics now has a live online store at haq1.com. The team lists its own products through the admin panel I built for them, and the shop is doing well.